



COMPUTER

- The process when processor fetch or decode another instruction during the execution of current instruction is called
(a) Supercomputing (b) Pipelining
(c) Cloud computing (d) Grid computing
- Which of the following is used by ALU to store the intermediate results?
(a) Stack (b) Heap
(c) Registers (d) Accumulators
- One TeraByte (TB) = GB and
One TeraByte (EB) = GB
(a) 2^{10} GB , 2^{16} GB (b) 2^{10} GB , 2^{20} GB
(c) 2^{10} GB , 2^{24} GB (d) 2^{10} GB , 2^{30} GB
- The cache memory is more effective because of
(a) Memory localization
(b) Locality of reference
(c) Memory size
(d) None of the mentioned
- Which of the following is the fastest means of memory access for CPU?
(a) Registers (b) cache
(c) Main memory (d) Stack
- The number $(2217)_8$ is equivalent to
(a) $(608)_{16}$ (b) $(028F)_{16}$ (c) $(048F)_{16}$ (d) $(2297)_{10}$
- To fetch data from the secondary memory which one of the following register is used?
(a) MAR (b) PC (c) IR (d) MBR
- The binary multiplication 00×11 will give
(a) 11 (b) 00 (c) 01 (d) 10
- Consider a computer system with speed of 10^6 instructions per second. A program P, having $2n^2$ steps is run on this system, where n is the input size. If $n = 10000$, what is the execution time for P?
(a) 1.2 (b) 20s (c) 100s (d) 200s
- To access the I/O devices, the status flag is continuously checked in
(a) Program controlled I/O (b) Memory mapped I/O
(c) I/O mapped (d) None of these

ANSWER

1.	B	2.	D	3.	D	4.	B	5.	A
6.	C	7.	A	8.	B	9.	D	10.	a

SOLUTION

- (b)
Pipe-lining in the process of improving the performance of the system by processing different instructions at the same time, with only one instruction performing one specific operation.
- (d)
Accumulators are used by ALU (Arithmetic Logic Unit) to store the intermediate results. The ALU is the computational center of the CPU. It performs all mathematical and logical operations.
- (d)
1 Giga Byte (GB) = $1024 = 10^{10}$ MB

$$\begin{aligned}
 1 \text{ Tera Byte (TB)} &= 2^{10} \text{ GB} \\
 1 \text{ Peta Byte (PB)} &= 2^{10} \text{ PB} \\
 1 \text{ ExaByte (EB)} &= 2^{10} \text{ PB} \\
 &= 2^{10} \times 2^{10} \text{ TB} \\
 &= 2^{10} \times 2^{10} \times 2^{10} \text{ GB} = 2^{30} \text{ GB}
 \end{aligned}$$

4. (b)

The cache memory is more effective because of locality of reference. This means that the cache depends on the locations in the memory that is referenced often.

5. (a)

Registers are fastest means of access for CPU. Registers are the small memory locations which are present closet to the CPU.

6. (c)

Binary equivalent of $(2217)_8$ is

$$\begin{array}{cccc}
 & 2 & & 2 & & 1 & & 7 \\
 & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 & 010 & & 010 & & 001 & & 111 \\
 & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 & 010010001111 & & & & & &
 \end{array}$$

$= (010010001111)_2$

Hexadecimal equivalent of $(010010001111)_2$ is

$$\begin{array}{ccc}
 0100 & 1000 & 1111 \\
 \downarrow & \downarrow & \downarrow \\
 4 & 8 & 15 \rightarrow F = (48F)_{16}
 \end{array}$$

$$\text{So } (2217)_8 = (048F)_{16}$$

7. (a)

To fetch data from the secondary memory, MAR (Memory Address Register) is used. MAR is a CPU register that either stores the memory address from which data will be fetched to the CPU or the address to which data will be sent and stored.

8. (b)

Binary multiplication for 00×11

$$\begin{array}{r}
 00 \\
 \times 11 \\
 \hline
 00 \\
 00 \\
 \hline
 000
 \end{array}$$

So, it will give output as 00.

9. (d)

Number of steps = $2n^2$
Speed of computer = 10^6 per second
For $n = 10000 (10^4)$

$$\begin{aligned}
 \therefore \text{Time} &= \frac{\text{Number of Steps}}{\text{Speed of Computer}} \\
 &= \frac{2n^2}{10^6} = \frac{2 \times (10^4)^2}{10^6} = \frac{2 \times 10^8}{10^6} \\
 &= 2 \times 10^2 = 200 \text{ sec.}
 \end{aligned}$$

10. (a)

To access the I/O devices, the status flag is continuously checked in program controlled I/O. In this method, the processor constantly checks the status flags and when it finds that the flag is set it performs the appropriate operation.

